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Jongho Kim

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Cornell Johnson Graduate School of Management — https://jonghkim.github.io

Education

Cornell University, SC Johnson Graduate School of Management — Aug 2021–Present
Ph.D. Candidate in Marketing
Committee: Vrinda Kadiyali (Chair), Emaad Manzoor, Omid Rafieian, Dan Kowal

Korea Advanced Institute of Science and Technology — Feb 2015–Feb 2017
M.S. in Management Engineering, Information Systems

Ulsan National Institute of Science and Technology — Mar 2010–Aug 2014
B.S. in Computer Science and Engineering, Minor in Finance and Accounting

Journal Publications

Park, J. and **Kim, J.**, 2022. "A Data-Driven Exploration of the Race between Human Labor and Machines in the 21st Century," *Communications of the ACM* [Manuscript] [SSRN] [Website] [Video] [Code]

Choi, S., Kim, W. and **Kim, J.**, 2016. "Compression of Hamiltonian Matrix: Application to Spin-1/2 Heisenberg Square," *AIP Advances* (**Editor's Pick**) [Manuscript]

Working Papers

Kim, J. and Kadiyali, V., "Estimating Treatment Effects with Large Language Models: Evidence from San Diego's Algorithmic Pricing Ban" **(Job Market Paper)**

Kim, J., Kowal, D., and Ruppert, D., "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors"

Kim, J., Anand, P., and Kadiyali, V., "The Effects of Rent Control on Renter Experience and Consumption"

Selected Conference and Seminar Presentations

Kim, J., Kowal, D., and Ruppert, D., 2026, "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors," *IO / Empirical Modeling Brown Bag*, Cornell University, Ithaca, NY, USA

Kim, J., Anand, P., and Kadiyali, V., 2025, "The Effects of Rent Control on Consumer Reviews on Rental Housing," *ISMS Marketing Science Conference*, Washington, D.C., USA

Park, J. and **Kim, J.**, 2018. “Fixing Racial Discrimination through Analytics on Online Platforms: A Neural Machine Translation Approach,” *International Conference on Information Systems (ICIS)*, San Francisco, California, USA [Manuscript] [Poster]

Kim, J. and Park, J., 2017. “Does Facial Expression Matter Even Online? An Empirical Analysis of Facial Expression of Emotion and Crowdfunding Success,” *International Conference on Information Systems (ICIS)*, Seoul, Korea [Manuscript] [Poster]

Kim, J., Lee, M., Cho, D. and Lee, B., 2016. “Are All Spillovers Created Equal? The Impact of Blockbusters and the Composition of Backers in Online Crowdfunding,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Presentation]

Kim, J., Cho, D. and Lee, B., 2016. “The Mind Behind Crowdfunding: An Empirical Study of Speech Emotion in Fundraising Success,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland (**Most Innovative Research-in-Progress Paper Runner-up**) [Manuscript] [Poster]

Park, J., **Kim, J.** and Lee, B., 2016. “Which Tasks Will Technology Take? A New Systematic Methodology to Measure Task Automation,” *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Poster]

Professional Experience

December & Company Asset Management, Seoul, South Korea — Apr 2019–Jul 2021
Senior Researcher / Quantitative Portfolio Manager, Portfolio Research Team

- Develop Generative Models for Textual Factors to Predict and Explain Asset Returns based on SEC-10X (10-K, 10-Q, 10-KSB and 10-QSB)
- Implement a Sector Rotation Strategy through Cross-Sectional Variation over Data-driven Economic Cycles
- Construct Alternative Investment Strategies, especially for Commodity, REITs, Infrastructure and High Dividend-Paying ETFs
- Work as a Substitute for Mandatory Military Service

NICE Pricing and Information, Seoul, South Korea — Aug 2016–Apr 2019
Quantitative Researcher, Financial Engineering Center

- Develop Non-Homogeneous Time Markov Chain Models (NHCTMC) for Forward Looking Probability of Default under IFRS9 based on Sparse Transition Matrix of Credit Ratings
- Estimate Maximum Likelihood Proxy Portfolio of Fixed-Income Portfolio via Return based Style Analysis (RBSA)
- Implement Implied Volatility Surface of Index Futures Options via Heston Stochastic Volatility Model
- Work as a Substitute for Mandatory Military Service

Digital Economics and Enterprise Research Lab, KAIST, Seoul, Korea — May 2015–Feb 2017

- Empirical Research in Information Systems using Machine Learning and Econometrics (Advisor: Dr. Byungtae Lee)
- Accepted Three Conference Papers in ICIS 2016 and Awarded for the Most Innovative Research-in-Progress Paper Award (Runner-Up) in ICIS 2016
- Government & Industry Projects

  Strategies for Online to Offline (O2O) Transport Services, The Korea Transport Institute (Advisor: Dr. Lee, B.), Sep 2016–Nov 2016

  The Effect of Model-Driven Development (MDD) on Information Systems Maintenance, LG CNS (Advisor: Dr. Lee, B.), Sep 2015–Dec 2015

Computational Mathematical Science Lab, UNIST, Ulsan, Korea — Sep 2011–Feb 2015

- Research and Study for Complex Network Theory and Numerical Methods for Differential Equations
- Published a Paper in AIP Advances (Compressed Representation of the Hamiltonian Matrix) and Selected as an Undergraduate Research Opportunity Program (Optimal Control Strategies of Lotka–Volterra Equation)

NewsJAM, Ulsan, South Korea — Jun 2013–Feb 2014

Chief Technology Officer (CTO), Co-Founder

- Developed NLP engines for news analysis including text summarization, key sentence extraction, topic modeling, and text clustering [PyCon Korea 2016]
- Open SW Incubator ($60,000 as award), National IT Industry Promotion Agency, 2013

Honors & Awards

AMA-Sheth Doctoral Consortium Fellow	2026
ISMS Doctoral Consortium Fellow	2025
FinTech@Cornell Student Fellowship	2022
Cornell SC Johnson College Doctoral Fellowship	2021–current
Best Student Paper Award, Post-ICIS 2017 KrAIS Research Workshop	2017
Best Graduate Student Paper Award, KAIST	2017
Most Innovative Research-in-Progress Paper Runner-Up, International Conference on Information Systems (ICIS), Dublin, Ireland	2016
Editor's Picks from the American Institute of Physics, AIP Advances	2016
Scholarship for Academic Excellence among 2015 Entering Class, KAIST	2015
Excellence Prize (Top 1%, 3rd Place over 322 teams) at Hackathon for Big Data, Ministry of Science, ICT and Future Planning	2014
Dean's List for Four Semesters, UNIST	2010, 2012–2014
Undergraduate Research Opportunity Program, Korea Foundation for Advancement of Science & Creativity	2013
National Natural Science and Engineering Scholarship, Korea Student Aid Foundation	2010–2014

Patents

Kim, J., et al., "Automated Trading System and Method," Korean Patent 10-2433324, Assignee: December & Company Inc., 2022. (Application No. 10-2019-0117628, filed Sep 24, 2019)

Teaching

Instructor

Cornell University — Fall 2025

BANA 5440 Introduction to Data Programming, MSBA

StudyPie — Oct 2018–Apr 2019

Deep Learning Models for Financial Time Series Prediction [Link]

Guest Lecturer

KAIST FMBA – Introduction to FinTech (Prof. Lee, B.) — Mar 2021, Mar 2023, Apr 2024

Investment Strategies in FinTech Industry [Link]

Korea Summer Session on Causal Inference (Prof. Park, J.) — Jul 2021

Applications of Machine Learning in Marketing [Link]

KAIST MBA – Business Analytics (Prof. Oh, W.) — Apr 2019

Intelligent Video Analytics with Deep Learning [Link]

KAIST IT Management Summer Research Workshop (Prof. Park, J.) — Aug 2017

Machine Learning APIs and Web Scraping [Link]

Teaching Assistant

Cornell University

NCCT 5030/5031 Marketing Management — Spring, Fall 2023

AEM 4435 / NBA 6390 Data-Driven Marketing — Fall 2022

NCC 5030 Marketing Management — Summer 2022

Invite-Only Workshops

Econometric Society, Africa Training Workshop in Econometrics, 2024

Services

Ad-hoc Reviewer

International Conference on Information Systems (ICIS), 2021

International Conference on Information Systems (ICIS), 2019

International Conference on Electronic Commerce (ICEC), 2017

Other Information

Work Authorization: Employment Authorization Document (EAD) Expected

Language: Korean (Native), English (Fluent)

Programming: Python, C/C++, Java, R, STATA, Matlab, SQL, JavaScript, HTML, CSS

References

Vrinda Kadiyali (Chair)
Nicholas H. Noyes Professor of Management
Professor of Marketing and Economics
SC Johnson Graduate School of Management
Cornell University
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Selected Research Abstracts

"Estimating Treatment Effects with Large Language Models: Evidence from San Diego's Algorithmic Pricing Ban," Kim, J. and Kadiyali, V. (Job Market Paper)

How can we use LLM outputs for causal inference when they are imperfect rather than ground truth? LLMs can measure constructs from text at scale, but they may invent mentions that are absent, omit ones that are present, and miscalibrate the values they report. We develop a Bayesian framework for causal inference with imperfect LLM measurements in panel data. With a small labeled sample and many LLM outputs, we estimate the treatment effect on satisfaction underlying discrete ratings, correcting the LLM's selection (invented or omitted mentions) and miscalibration while accounting for labeling uncertainty. We apply our framework to San Diego's 2025 algorithmic-pricing ban, which prohibits setting rents from competitors' nonpublic data, using 58,293 Google Maps reviews of 1,686 apartments across the city and comparable control cities. In the nine months after the ban, renters rate their apartments 0.15 points lower on a 5-point scale, counter to its pro-renter intent. From the review text, we find satisfaction with rent price falls by 0.26 while satisfaction with non-price services rises by 0.37. The ban appears to raise pricing inefficiency and shift landlord competition from price to non-price services, leaving renters worse off. As the DOJ–RealPage settlement extends the same regime nationwide, our findings inform how policymakers weigh its full consequences for renter satisfaction.

"The Effects of Rent Control on Renter Experience and Consumption: Evidence from St. Paul's 2022 Ordinance," Kim, J., Anand, P., and Kadiyali, V.

How does housing affordability policy reshape consumer experience and grocery spending? We examine the impact of St. Paul's 2022 rent control policy using 110,258 renter reviews from 2,285 apartments and grocery sales data from 123 stores in 18 months post-policy. We compare St. Paul (policy area) and surrounding spillover areas to comparable U.S. cities as controls. Renters rate their apartment experiences higher by 0.474 points (on a 5-point scale) in the policy area and by 0.157 points in the spillover areas. In the 18 months after policy implementation, no dimension of the multi-attribute service experience worsens, suggesting net satisfaction gains for renters in our time window. Grocery expenditure declines by 8% in the policy area and 3% in the spillover areas over the two years following implementation through different mechanisms: the exit of higher-income renters from the policy area and likely increased rents in the spillover areas. This reveals a policy tradeoff: improved renter satisfaction dampened grocery demand in affected neighborhoods. Our findings have implications for apartment managers to optimize service quality and consumer retention under price regulation; grocery retailers to recalibrate assortment and pricing for shifting neighborhood demographics; and policymakers to weigh the full consequences of rent control across renters, businesses, and spillover areas.

“Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors,” Kim, J., Kowal, D., and Ruppert, D.

Large language models (LLMs) enable researchers to measure latent constructs from text at scale, but their use in statistical inference raises challenges when measurement errors are correlated. We study treatment effect estimation when outcomes are measured using LLM outputs calibrated with a limited validation sample. We show that standard plug-in estimators and regression calibration can substantially underestimate uncertainty—even when point estimates are unbiased—because they fail to account for correlated measurement errors and calibration uncertainty. Using data from a randomized experiment, we empirically document that LLM measurement errors exhibit systematic dependence structured by embedding-space similarity, which persists even after fine-tuning and renders standard clustering approaches inadequate. To address this problem, we develop a Bayesian framework that jointly models calibration uncertainty and correlated measurement error by learning the error covariance structure from embedding geometry. Our approach provides coherent uncertainty quantification and clarifies when efficiency gains from LLM-based measurement are achievable.